



A Physics-Informed Fractional Framework for Modeling Post-Earthquake Disaster Risk, Public Health Burden, and Recovery Dynamics

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Abstract: Earthquakes can cause interconnected disruptions to the built environment, population health, emergency response systems, and community recovery, with consequences that may persist long after the initial seismic event. These post-earthquake dynamics are inherently time-dependent and may exhibit memory and hereditary effects that are not adequately represented by conventional integer-order differential equations. A physics-informed fractional framework was therefore developed to characterize the coupled evolution of post-earthquake disaster impacts and recovery processes. Five state variables were introduced to represent physical damage, population health burden, shelter demand, emergency response capacity, and community recovery. Their interactions were described through a coupled system of fractional-order differential equations. A physics-informed neural network was subsequently formulated by embedding the governing fractional-order equations and initial-condition constraints directly into the learning objective. Reference numerical trajectories were generated using the fractional Adams–Bashforth–Moulton method to assess the internal numerical consistency of the scenario-based simulations. Synthetic datasets informed by publicly available earthquake-related indicators were used to examine the computational behavior of the proposed framework under representative post-earthquake scenarios. The resulting formulation provides an integrated computational representation of the temporal dependencies among disaster impacts, public health burden, emergency response capacity, shelter demand, and community recovery while explicitly accounting for memory effects through fractional-order dynamics. The framework establishes a methodological basis for investigating post-earthquake system evolution and for examining how persistent disaster effects may influence recovery trajectories. Following validation against empirical observations from real earthquake events, the proposed approach could support scenario analysis, disaster preparedness, public health planning, emergency resource allocation, and quantitative assessment of post-earthquake recovery.

Keywords: Earthquakes; Natural disasters; Public health burden; Fractional differential equations; Physics-informed neural networks; Disaster recovery

1 Introduction

Earthquakes are among the most destructive of the natural disasters that impact the planet. Beyond the loss of life and injuries that can occur during earthquake events, there are often additional consequences of major earthquakes, such as overcrowding in hospitals, displaced populations, health issues related to the sheltering of displaced individuals, and even socioeconomic issues that result from earthquakes [1–4]. The earthquakes that occurred in Türkiye and Syria in 2023 show that the impact of earthquakes extends beyond evaluating the damage to the structures and the loss of life that results from those earthquakes; the earthquakes demonstrated the potential for prolonged and interconnected crises involving health, shelter, infrastructure, and emergency and recovery needs [1–5].

Türkiye is located in one of the most seismically active regions on the planet and has experienced several earthquakes that have led to significant social and public health consequences. The earthquakes, which struck the southeastern region of Türkiye and northern Syria on February 6, 2023, caused considerable destruction and

significant needs within the affected areas [2, 5]. The emergency medical services were overwhelmed by the number of patients that required immediate treatment. Essential health services were disrupted following the earthquake and other disasters caused by the earthquake. There were critical needs for food, water, shelter, and health care for these affected populations [2, 5]. The earthquake not only created significant damage to the area but also presented a great opportunity to develop a model that would aid in the treatment of these affected individuals. Thus, there was a need for models to encompass these concepts and assist in the recovery of the individuals impacted by the earthquake.

In understanding the field of disaster management, it is essential to gain an understanding of each of these elements and how they interact with each other. The Sendai Framework for Disaster Risk Reduction 2015–2030 has established four major priorities for the global reduction of disasters and their impact on human populations: understanding disaster risk, strengthening disaster risk governance, investing in resilience, and enhancing preparedness for effective response and “Build Back Better” recovery [6]. However, within these frameworks for disaster risk management, the health-related impacts of such disasters should also be considered and addressed. These aspects include public health preparedness, health system resilience, and health-related recovery after the disaster occurs [7].

Following an earthquake, the recovery process is not linear with time. While earthquake damage to the infrastructure can decrease over time due to repair and reconstruction efforts, various factors influence the rate of recovery of the earthquake-damaged area, such as the emergency response, healthcare, and shelter facilities for the displaced individuals. Following an earthquake, the public health burden continues to be high due to earthquake injuries, the impacts of disrupted chronic illnesses, poor sanitation in shelters, communicable illnesses, mental health issues, and a lack of access to healthcare facilities [1, 3, 4, 8]. Thus, the model should include variables related to the physical damage to the area following the earthquake as well as the public health burden, shelter needs, response capacity, and the rate of recovery of the affected area.

Various models have been developed to characterize disaster recovery processes. However, the majority of these models use static indicators, statistical measures, or integer-order differential equations. Each of these options has limitations when it comes to accurately modeling certain aspects of disaster models. One alternative is the use of fractional differential equations, which can account for the memory properties of the systems in question [9]. Systems related to disaster recovery can benefit significantly from the ability of fractional calculus to model the impact of previous damage and the effects of delayed responses to that damage.

At the same time, another group of artificial intelligence methods that has recently emerged as helpful for solving differential equations and learning about dynamic systems is physics-informed neural networks [10]. Rather than requiring data from a system to be trained, these models can incorporate information regarding the differential equations that describe the system into the loss function of the network, enabling it to learn from both the data from the system and the equations that describe it [10]. This is especially helpful for problems related to natural disasters, where data from the systems being modeled may not be available. Furthermore, recent surveys of the field have suggested that physics-informed methods may be especially helpful for modeling systems with differential equations and scientific knowledge [11].

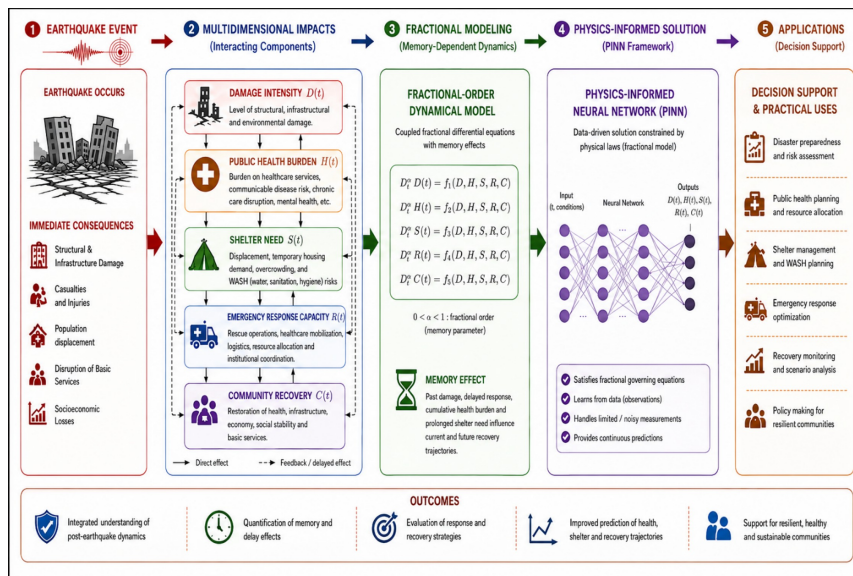


Figure 1. Overall graphical overview of the proposed physics-informed fractional framework

Note: PINN = Physics-informed neural network; WASH = water, sanitation, and hygiene.

Despite the growing interest in fractional modeling and physics-informed learning, there is a lack of unified

frameworks that consider the various physical and health-related aspects of earthquakes and the post-earthquake disaster. To address this gap in the literature, a framework based on physics-informed fractional calculus is proposed for modeling earthquake disasters, public health, and the recovery of the affected communities. The model considers five variables that describe the physical and health aspects of earthquakes and their aftermath: damage, public health, shelter, emergency response, and recovery. The results of this model can aid in the planning of effective disaster response and recovery strategies for earthquake-prone areas.

Figure 1 summarizes how earthquake-induced damage, public health burden, shelter need, emergency response capacity, and community recovery are integrated into a unified fractional and physics-informed computational framework for post-earthquake disaster recovery analysis.

2 Novelty and Contribution

While the topic of earthquake recovery has been extensively studied from a variety of different perspectives, most existing studies have focused on individual components of the recovery process following an earthquake. For example, different studies have investigated the damage that occurs to structures following an earthquake, the recovery of those structures, and the logistics that must occur following those earthquakes, such as in the areas of healthcare and shelter [12–16]. Furthermore, various studies have applied physical neural networks to tasks like modeling seismic waves, tsunamis, landslides, and the seismic response of structural components [17–19]. Recent studies have increasingly emphasized the integration of computational intelligence, resilience assessment, and data-driven disaster recovery modeling to improve post-disaster decision-making and emergency preparedness [20–22]. However, these different studies have mainly been focused on investigating the physical components of earthquake recovery.

Fractional differential equations have been increasingly used to model systems that exhibit memory or delay effects [9]. Post-earthquake disasters exhibit such characteristics, as the current state of a community is not only dependent upon the current level of damage to its infrastructure but also upon the damage that has occurred in the past. To the best of our knowledge, no prior study has proposed a unified fractional-order framework that simultaneously integrates damage intensity, public health burden, shelter demand, response capacity, and recovery dynamics in post-earthquake settings.

The novelty of this study, therefore, does not lie in the fractional calculus or the physics-informed neural network methods used individually but rather in the use of these methods together within the proposed disaster recovery model. The model utilizes fractional calculus to extend current models of disaster recovery by accounting for memory-dependent, nonlinear, and health-sensitive aspects of the recovery following an earthquake. These aspects relate to the Sendai Framework for Disaster Risk Reduction, which aims to understand disaster risk, increase resiliency, and prepare for effective responses and “Build Back Better” recovery efforts following a given disaster [6].

Figure 2 illustrates how previous studies have generally addressed disaster modeling, public health consequences, fractional dynamics, and physics-informed learning as separate research streams. The proposed study contributes by integrating these components into a unified post-earthquake recovery framework.

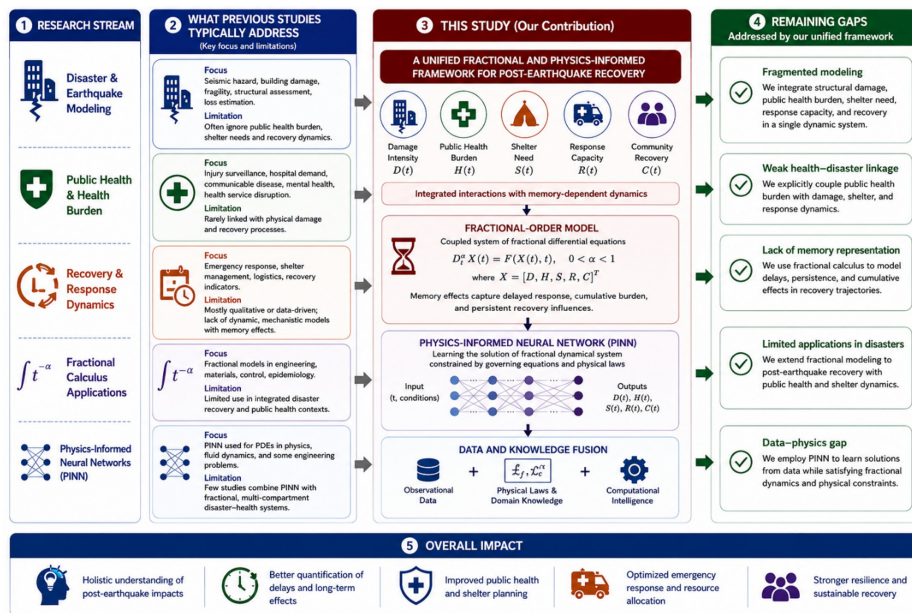


Figure 2. Research gap and novelty map of the proposed study

Note: PDE = partial differential equation.

The main contributions of this study are as follows:

- (i) A new fractional-order post-earthquake recovery model is formulated to represent memory-dependent disaster dynamics.
- (ii) The public health burden is considered to be one of the main state variables of the system.
- (iii) The variables that relate to shelter needs, the response to the shelter needs, the damage that is performed upon the individuals in the shelter, and the recovery of those individuals from the system are all considered to be components of the same system.
- (iv) A physics-informed neural network formulation is incorporated as a computational component of the framework by embedding the governing fractional equations and initial-condition constraints into the learning objective; the Adams–Bashforth–Moulton solution is used as the numerical reference for the scenario-based fractional dynamics.
- (v) Various scenarios are considered within the study to determine the impact that different variables have upon the public health and the recovery of the individuals in the system.
- (vi) The proposed framework provides a computational basis for disaster preparedness, public health planning, emergency resource allocation, and sustainable post-earthquake recovery strategies.

3 Aim of the Study

The aim of this study is to develop a physics-informed fractional computational framework for modeling post-earthquake disaster risk, public health burden, shelter need, emergency response capacity, and community recovery dynamics.

Specifically, this study aims to:

- (i) formulate a fractional-order dynamic model that captures memory-dependent post-earthquake recovery processes;
- (ii) integrate public health burden as a central component of disaster recovery rather than as a secondary outcome;
- (iii) model the interactions among infrastructure damage, shelter need, emergency response capacity, and community recovery;
- (iv) incorporate a physics-informed neural network formulation as a computational component for approximating the proposed fractional system;
- (v) evaluate different post-earthquake scenarios using synthetic scenario-based data inspired by publicly reported disaster indicators;
- (vi) provide a computational framework that may support disaster preparedness, public health planning, emergency resource allocation, and sustainable recovery strategies.

This integrated framework aims to bridge the gap between disaster modeling, public health systems, and computational intelligence.

4 Conceptual Framework

Post-earthquake recovery is conceptualized as a coupled public health and disaster-management process. The proposed framework includes five interacting state variables: damage intensity $D(t)$, public health burden $H(t)$, shelter need $S(t)$, emergency response capacity $R(t)$, and community recovery $C(t)$. Damage intensity refers to the level of damage that the earthquake has caused to the environment and its structures. The public health burden refers to the impact of the earthquake on emergency rooms, hospitals, healthcare workers, and communicable and non-communicable diseases. The shelter need pertains to the number of people that need to be housed following the earthquake. The earthquake's emergency response refers to the number of emergency responders, healthcare providers, and the resources required to respond to the earthquake. Finally, the concept of community recovery refers to the recovery of the area from the earthquake over time.

The conceptual assumption of the model is that earthquake damage affects both the public health and the need for shelter. The emergency response to earthquake damage can mitigate the impact of that damage upon the public health and the need for shelter, but high levels of both of those parameters following an earthquake negatively affect response capacity. In recognizing that the recovery period following an earthquake is not a memoryless process, the model can incorporate the effects of previous damage and the need for recovery into the calculations of the subsequent recovery period. Thus, fractional-order terms can be used to model such memory-dependent recovery dynamics.

The framework in Figure 3 illustrates the interactions among five key components: $D(t)$, $H(t)$, $S(t)$, $R(t)$, and $C(t)$. Solid arrows denote direct influences, while dashed arrows represent feedback and delayed effects; positive (+) and negative (−) signs indicate the direction of influence on the affected component.

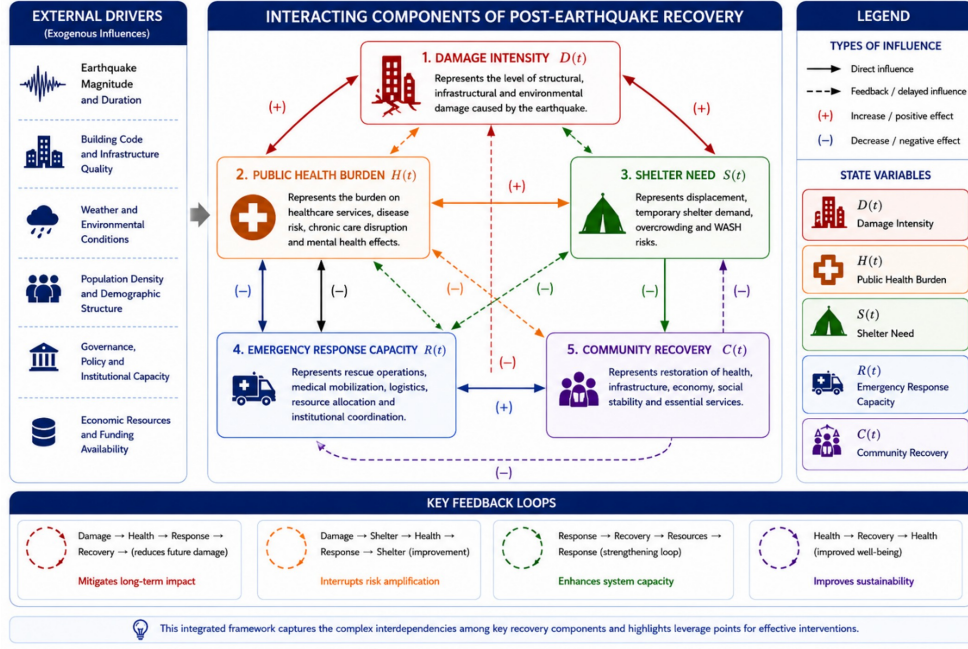


Figure 3. Conceptual framework of post-earthquake public health recovery dynamics

Note: WASH = water, sanitation, and hygiene.

5 Proposed Fractional Mathematical Model

Let t denote the time elapsed since the earthquake. The proposed model is formulated as a coupled fractional-order system involving $D(t)$, $H(t)$, $S(t)$, $R(t)$, and $C(t)$. The system is defined as follows:

$$D_t^\alpha D(t) = \lambda_E(1 - D(t)) - \beta_R R(t)D(t) - \mu_D D(t) \quad (1)$$

$$D_t^\alpha H(t) = \eta_D D(t) + \eta_S S(t) - \beta_H R(t)H(t) - \mu_H H(t) \quad (2)$$

$$D_t^\alpha S(t) = \rho_D D(t) - \beta_S R(t)S(t) - \mu_S S(t) \quad (3)$$

$$D_t^\alpha R(t) = \sigma_0 + \sigma_C C(t) - \delta_R R(t) - \omega_H H(t) \quad (4)$$

$$D_t^\alpha C(t) = \gamma_R R(t)(1 - C(t)) - \gamma_D D(t)C(t) - \gamma_H H(t)C(t) \quad (5)$$

where, $0 < \alpha \leq 1$ denotes the fractional order. When $\alpha = 1$, the system reduces to a classical integer-order model. When $0 < \alpha < 1$, the system incorporates memory effects and represents delayed or cumulative recovery behavior.

Figure 4 illustrates the coupled fractional-order system describing the interactions among $D(t)$, $H(t)$, $S(t)$, $R(t)$, and $C(t)$. The Caputo fractional derivative is employed to capture memory-dependent effects, allowing past damage, delayed response, and accumulated health burden to influence future recovery trajectories.

Damage intensity within Eq. (1) increases as a result of earthquake damage, yet it decreases with response and repair processes. Health burden increases with earthquake damage and shelter needs, yet decreases with the response to those damages and needs represented in Eq. (2). Shelter need increases with earthquake damage but decreases with response and reconstruction efforts represented in Eq. (3). Response capacity increases with baseline and recovery of affected communities but decreases with exhaustion of those response efforts and public health systems represented in Eq. (4). Finally, community recovery increases with response capacity but decreases with earthquake damage and health burden represented in Eq. (5). The fractional system described in this study allows for modeling of disaster recovery as a process that can be represented by a system of nonlinear mathematical equations that consider the impact of various factors on the recovery of communities after earthquake damage. Table 1 shows the model parameters and their descriptions.

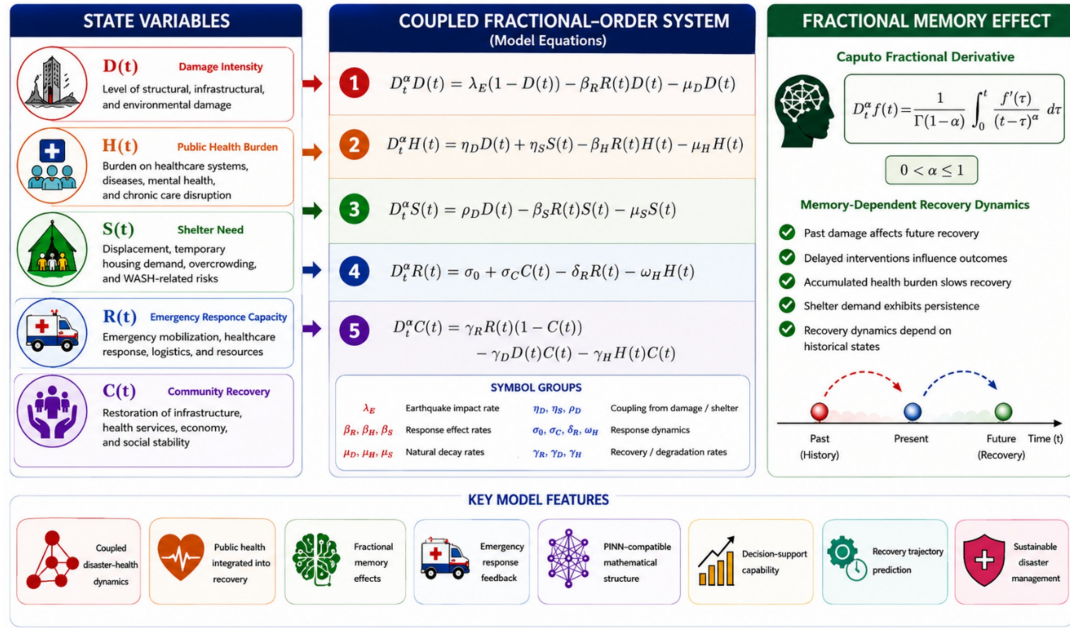


Figure 4. Structure of the proposed fractional mathematical model for post-earthquake recovery dynamics
Note: WASH-related risks = risks related to water, sanitation, and hygiene; PINN = Physics-informed neural network.

Table 1. Model parameters and their descriptions

Parameter	Description
λ_E	Earthquake-induced damage activation rate
β_R	Effect of response capacity on damage reduction
μ_D	Natural repair or damage reduction rate
η_D	Effect of damage on public health burden
η_S	Effect of shelter need on public health burden
β_H	Effect of response capacity on reducing health burden
μ_H	Natural reduction rate of health burden
ρ_D	Effect of damage on shelter need
β_S	Effect of response capacity on reducing shelter need
μ_S	Natural reduction rate of shelter need
σ_0	Baseline emergency response mobilization
σ_C	Effect of community recovery on response capacity
δ_R	Exhaustion or decay of response capacity
ω_H	Negative effect of health burden on response capacity
γ_R	Recovery acceleration due to response capacity
γ_D	Negative effect of damage on recovery
γ_H	Negative effect of health burden on recovery

6 Physics-Informed Neural Network Framework

Recent developments have demonstrated the effectiveness of physics-informed neural networks for solving scientific machine learning problems involving ordinary and partial differential equations, inverse problems, and data-scarce systems [23–26]. In the present study, the term “physics” refers to the governing system dynamics and mathematical constraints represented by the proposed fractional differential equations. It does not imply immutable conservation laws such as mass or energy conservation; rather, it refers to embedding domain-specific disaster-recovery dynamics into the neural network training process. To solve the proposed fractional-order system and learn the underlying dynamics of post-earthquake disaster recovery, a physics-informed neural network is employed. Unlike purely data-driven models, physics-informed neural networks incorporate the governing equations of the system into the learning process, enabling the model to respect physical and mathematical constraints while fitting available data.

In this study, the neural network takes time t as input and predicts the state variables:

$$\hat{Y}(t) = [\hat{D}(t), \hat{H}(t), \hat{S}(t), \hat{R}(t), \hat{C}(t)] \quad (6)$$

where, $D(t)$, $H(t)$, $S(t)$, $R(t)$, and $C(t)$ denote the predicted damage intensity, public health burden, shelter need, emergency response capacity, and community recovery, respectively.

6.1 Network Architecture

The physics-informed neural network consists of a fully connected feed-forward neural network. The input layer contains one neuron associated with time t . There are four hidden layers, each containing 50 neurons with hyperbolic tangent (tanh) activation functions. Finally, the output layer contains five neurons associated with the variables to be predicted: $D(t)$, $H(t)$, $S(t)$, $R(t)$, and $C(t)$. The neural network is trained to learn the solution to the fractional differential equation system rather than simply finding the best fit to the raw data. The Adam optimizer with a learning rate of 0.001 is used for training the network for 5000 epochs.

6.2 Loss Function

The total loss function is defined as a combination of three components:

$$L = L_{\text{data}} + L_{\text{physics}} + L_{IC}$$

where, L_{data} measures the difference between predicted values and scenario-based data; L_{physics} enforces the fractional differential equations; and L_{IC} ensures consistency with initial conditions. The physics-based loss is computed by substituting the neural network outputs into the governing fractional equations and minimizing the residuals.

6.3 Fractional Constraint Integration

To incorporate fractional dynamics, the Caputo fractional derivative is used:

$$D_t^\alpha x(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{x}(\tau)}{(t-\tau)^\alpha} d\tau$$

This formulation allows the model to capture memory-dependent dynamics, where past states influence present behavior.

6.4 Advantages of the Physics-Informed Neural Network Approach

The use of physics-informed neural networks provides several advantages:

- Ability to learn from limited or incomplete data;
- Incorporation of physical laws and system equations;
- Robustness to noisy observations;
- Continuous-time predictions;
- Compatibility with fractional-order systems.

Thus, the physics-informed neural network framework enables a hybrid modeling approach that integrates data, mathematical structure, and physical interpretation for post-earthquake disaster recovery.

7 Scenario-Based Simulation and Experimental Design

To evaluate the proposed model, scenario-based simulations were performed. Due to the limited availability of complete and harmonized post-earthquake datasets, controlled synthetic scenarios were constructed using normalized initial conditions and parameter ranges informed by publicly reported earthquake-impact indicators and previous disaster-recovery studies.

7.1 Synthetic Dataset Construction

To construct the scenario-based dataset, the indicators of earthquake damage that were reported publicly were grouped according to the state variables of the model. Indicators representing damage to structural elements in the affected area were incorporated into the damage intensity variable, $D(t)$. Healthcare-related indicators were incorporated into the public health burden variable, $H(t)$. Indicators related to the number of individuals to be housed in shelters were incorporated into the shelter need variable, $S(t)$. Indicators related to the ability of emergency responders to mitigate the earthquake's damage were incorporated into the response capacity variable, $R(t)$. Finally, indicators related to the restoration of the area after the earthquake were incorporated into the community recovery variable, $C(t)$. Each of these publicly reported indicators had different units and scales for their measurements. Thus,

they could not be directly comparable to each other in their raw form. Each of the state variables was normalized to a range of 0 to 1, with values close to 0 indicating low levels and values close to 1 indicating high levels.

For a generic indicator x , min–max normalization was defined as:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad x_{\text{norm}} \in [0, 1]$$

where, x represents the original indicator value, $x_{\min}$ represents the minimum value of the corresponding indicator, $x_{\max}$ represents its maximum value, and x_{norm} represents the normalized value.

The normalized representation was subsequently used to establish the initial conditions for the four representative post-earthquake scenarios. Scenarios S1 and S2 were assigned the same high initial damage intensity, $D(0) = 0.90$, but different initial emergency response capacities. Specifically, $R(0) = 0.20$ was used for S1 and $R(0) = 0.45$ for S2. This configuration allowed the influence of emergency response capacity to be examined under comparable initial damage conditions. Scenario S3 was designed to represent moderate structural damage accompanied by a comparatively high initial public health burden, with $D(0) = 0.65$ and $H(0) = 0.75$. Scenario S4 represented stronger response and accelerated recovery conditions, with $R(0) = 0.60$ and $C(0) = 0.25$.

The complete initial conditions for the four scenarios (S1–S4) are as follows:

$$\begin{aligned} \text{S1 : } & (D(0), H(0), S(0), R(0), C(0)) = (0.90, 0.45, 0.75, 0.20, 0.10), \\ \text{S2 : } & (D(0), H(0), S(0), R(0), C(0)) = (0.90, 0.45, 0.75, 0.45, 0.15), \\ \text{S3 : } & (D(0), H(0), S(0), R(0), C(0)) = (0.65, 0.75, 0.60, 0.30, 0.15), \\ \text{S4 : } & (D(0), H(0), S(0), R(0), C(0)) = (0.70, 0.40, 0.55, 0.60, 0.25). \end{aligned}$$

For each scenario, the parameter values were selected within the ranges specified in Table 2. For scenario S1, it is characterized by high damage to the earthquake’s infrastructure yet with limited capacity to respond to the damage that occurred. Scenario S2 is characterized by high damage yet with the ability to quickly respond to the earthquake. Scenario S3 is characterized by moderate structural damage accompanied by a comparatively high public health burden. Finally, Scenario S4 is characterized by high damage yet with high response capacity to the earthquake. Therefore, the synthetic dataset described in this document was constructed to represent the conditions after earthquakes based upon published literature, rather than based upon a specific historical earthquake dataset. The dataset can be used to compare the various scenarios of earthquake response conditions.

Table 2. Parameter ranges used in scenario simulations

Parameter	Range	Interpretation
λ_E	0.05–0.20	Residual earthquake-induced damage activation rate
β_R	0.20–0.60	Response effect on damage reduction
μ_D	0.05–0.20	Natural repair rate
η_D	0.20–0.50	Effect of damage on health burden
η_S	0.10–0.40	Effect of shelter on health burden
β_H	0.20–0.60	Response effect on health burden
μ_H	0.05–0.20	Natural recovery rate
ρ_D	0.20–0.50	Effect of damage on shelter
β_S	0.20–0.60	Response effect on shelter
μ_S	0.05–0.20	Shelter recovery rate
σ_0	0.10–0.30	Baseline response
σ_C	0.10–0.40	Recovery impact on response
δ_R	0.05–0.20	Response exhaustion
ω_H	0.05–0.30	Health burden impact on response
γ_R	0.20–0.70	Recovery acceleration
γ_D	0.10–0.40	Damage effect on recovery
γ_H	0.10–0.40	Health effect on recovery

7.2 Initial Conditions

The initial state of the system is defined as:

$$D(0) = D_0, \quad H(0) = H_0, \quad S(0) = S_0, \quad R(0) = R_0, \quad C(0) = C_0$$

where, each variable is normalized within the range [0,1].

7.3 Simulation Scenarios

Four main scenarios were considered: Scenario S1 (high damage and low response capacity) represents a worst-case situation where emergency response is insufficient; Scenario S2 (high damage and rapid response) represents effective emergency mobilization and coordinated response; Scenario S3 (moderate damage and high public health burden) represents situations where healthcare systems are overwhelmed despite moderate structural damage; and Scenario S4 (strong response and accelerated recovery) represents optimal intervention strategies with efficient resource allocation.

7.4 Evaluation Metrics

The performance of the model was evaluated based on:

- Reduction in damage intensity over time;
- Peak and duration of public health burden;
- Duration of shelter need;
- Growth rate of community recovery;
- Stability and convergence of the system.

7.5 Simulation Objective

The objective of the simulations is to:

- understand how different response capacities affect recovery;
- analyze the interaction between health burden and infrastructure damage;
- evaluate the impact of fractional memory on recovery dynamics;
- provide insights for disaster preparedness and planning.

The scenario-based framework can be used to evaluate various response strategies to earthquakes and to aid in decision-making in the recovery from these earthquakes.

8 Numerical Results and Scenario Analysis

Simulations were performed to evaluate the proposed fractional-order model. The fractional differential equation system was numerically evaluated using the scenario-based computational framework, while the physics-informed neural network formulation was retained as a computational component for approximating the governing system dynamics. All variables were normalized to the range [0,1] and evaluated over the 100-day period following the earthquake.

8.1 Scenario Definitions and Initial Conditions

As shown in Table 3, Scenario S1 depicts a situation with a delayed or insufficient response to the disaster; Scenario S2 depicts an effective response to the disaster; Scenario S3 depicts a situation with a high health burden resulting from the disaster; and Scenario S4 depicts conditions that are optimal for recovery from the disaster.

Table 3. Scenario descriptions and initial conditions

Scenario	Description	$D(0)$	$H(0)$	$S(0)$	$R(0)$	$C(0)$
S1	High damage, low response capacity	0.90	0.45	0.75	0.20	0.10
S2	High damage, rapid response	0.90	0.45	0.75	0.45	0.15
S3	Moderate damage, high public health burden	0.65	0.75	0.60	0.30	0.15
S4	Strong response and accelerated recovery	0.70	0.40	0.55	0.60	0.25

Note: $D(0)$ = initial damage intensity; $H(0)$ = initial public health burden; $S(0)$ = initial shelter need; $R(0)$ = initial emergency response capacity; $C(0)$ = initial community recovery.

8.2 Parameter Ranges Used in Simulations

These parameter ranges were selected to represent contrasting low, moderate, and high response conditions within a controlled synthetic simulation environment. The range of the residual damage-activation parameter λ_E was revised to 0.05–0.20 after rechecking Eq. (1) against the normalized damage trajectories. Because λ_E acts as a persistent source term through $\lambda_E[1 - D(t)]$, substantially large values would maintain unrealistically high damage levels during the recovery period and would be inconsistent with the final normalized damage values obtained in the numerical scenarios. The revised range allows the damage-reduction terms associated with emergency response and natural repair to dominate progressively during the post-earthquake recovery phase while preserving a residual damage contribution.

Table 2 defines the range within which the parameters were controlled during the construction of the earthquake scenarios. The parameters were varied within these ranges according to the characteristics of each scenario (not according to the data from the earthquakes themselves). Scenarios S1 and S2 were constructed to represent situations with a high level of initial damage yet a limited and rapid response to that damage. Scenario S3 represents moderate structural damage accompanied by a comparatively high public health burden, whereas Scenario S4 represents stronger response capacity and accelerated recovery conditions.

Each of the state variables is normalized to the range $[0,1]$. Thus, the coefficients listed within the table represent the relative strengths of the variables within the model; the coefficients will need to be calibrated according to actual data from earthquakes and their effects in any future applications of the scenario model. The parameter selection followed a scenario-based rather than an empirical calibration strategy. Values were chosen within the predefined ranges to generate internally consistent contrasts among the four synthetic scenarios while preserving the qualitative roles of the interaction terms in Eqs. (1)–(5). Parameters associated with response, damage reduction, health-burden mitigation, shelter reduction, and recovery were therefore adjusted according to the defining characteristics of each scenario. These values should be interpreted as normalized relative interaction strengths for computational comparison rather than as earthquake-specific estimates.

The scenario-specific parameter values used in the numerical simulations are reported in Table 4. All parameter values lie within the predefined ranges in Table 2, including the revised range of the residual damage-activation parameter λ_E . The parameter values were chosen to preserve the conceptual characteristics of the four scenarios, with the goal of ensuring mathematical consistency with the fractional system that governs the scenarios. Scenario S1 represents a scenario with limited response and recovery. Scenario S2 represents a scenario with high initial damage and rapid response. Scenario S3 represents moderate structural damage accompanied by a comparatively high and persistent public health burden. Finally, Scenario S4 represents a scenario with both high response and recovery characteristics. These scenarios were defined within the fractional system as part of the numerical analysis of the system but do not represent the earthquake-specific coefficients.

Table 4. Parameter values for the four numerical simulation scenarios (S1–S4)

Parameter	S1	S2	S3	S4
λ_E	0.149	0.145	0.141	0.094
β_R	0.342	0.511	0.454	0.524
μ_D	0.177	0.152	0.074	0.131
η_D	0.231	0.284	0.312	0.273
η_S	0.148	0.154	0.197	0.192
β_H	0.443	0.403	0.566	0.342
μ_H	0.174	0.143	0.137	0.168
ρ_D	0.347	0.302	0.463	0.334
β_S	0.406	0.235	0.516	0.314
μ_S	0.159	0.141	0.158	0.053
σ_0	0.107	0.102	0.116	0.111
σ_C	0.100	0.109	0.110	0.103
δ_R	0.194	0.198	0.197	0.195
ω_H	0.128	0.125	0.102	0.185
γ_R	0.403	0.444	0.382	0.685
γ_D	0.130	0.131	0.122	0.100
γ_H	0.177	0.201	0.212	0.294

Note: S1–S4 denote the four simulation scenarios defined by damage intensity $D(t)$, public health burden $H(t)$, shelter need $S(t)$, emergency response capacity $R(t)$, and community recovery $C(t)$.

8.3 Numerical Simulation of Disaster Dynamics

Figure 5 shows the numerical evolution of the main system variables under Scenario S2. The damage intensity variable decreases over time as the emergency response and repair efforts become effective. The public health burden variable is high after the earthquake and decreases over time as the public health needs are met by the emergency response efforts. The shelter need variable increases at first due to the growing need for shelter for earthquake victims, then decreases over time as the shelter efforts become effective. Finally, the community recovery variable increases over time as the various necessary components of the community begin to be restored.

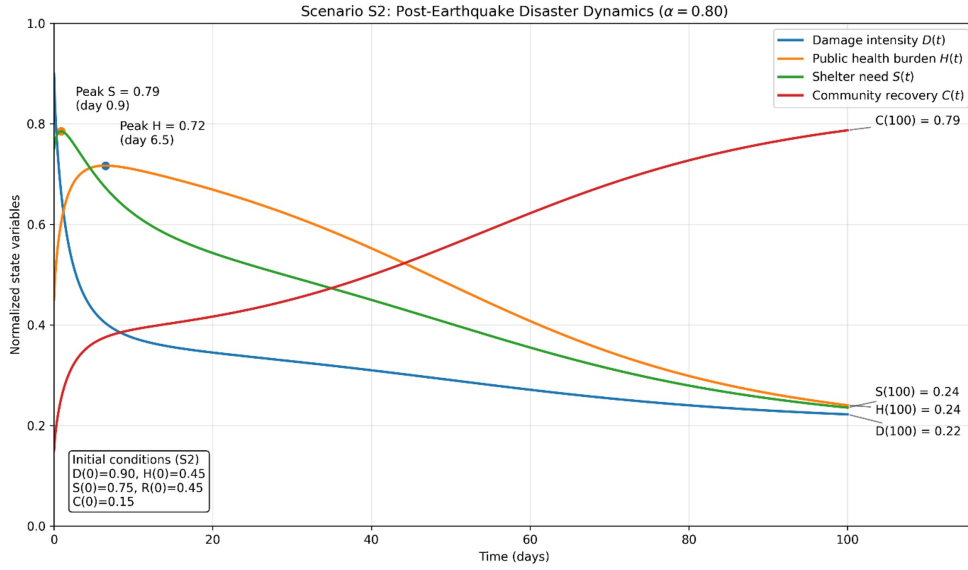


Figure 5. Numerical simulation of post-earthquake disaster dynamics under Scenario S2

Figure 5 shows the temporal evolution of $D(t)$, $H(t)$, $S(t)$, and $C(t)$. Damage intensity decreases over time, while shelter need exhibits an initial transient increase followed by a gradual decline. Public health burden is elevated during the early recovery phase and subsequently decreases, whereas community recovery progressively increases over the 100-day simulation period.

8.4 Effect of Fractional Order on Recovery

Figure 6 illustrates the effect of the fractional-order parameter α on community recovery under the S2 baseline scenario. The initial conditions were fixed at $D(0) = 0.90$, $H(0) = 0.45$, $S(0) = 0.75$, $R(0) = 0.45$, and $C(0) = 0.15$, and all model parameters were held constant at the Scenario S2 values reported in Table 4. Only the fractional-order was varied from $\alpha = 0.60$ to $\alpha = 1.00$. The results show that lower values of α produce slower recovery trajectories, consistent with stronger memory effects. At day 100, the community recovery level increased from $C(100) = 0.483$ for $\alpha = 0.60$ to $C(100) = 0.874$ for the classical case $\alpha = 1.00$.

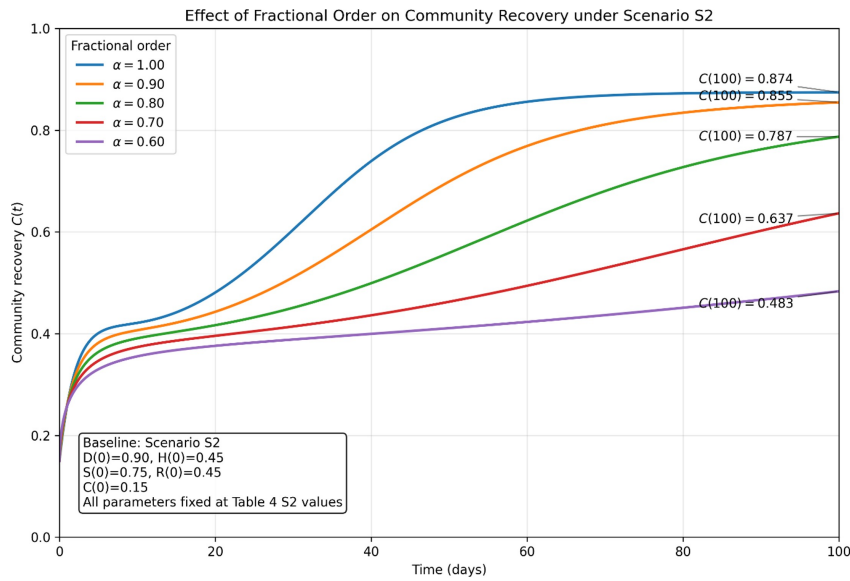


Figure 6. Effect of the fractional order on community recovery under the Scenario S2 baseline configuration

All initial conditions and model parameters were fixed at the S2 values reported in Table 3 and Table 4, while only α was varied from “0.60” to “1.00”. Lower fractional orders produce slower recovery because of stronger memory effects.

8.5 Numerical Verification Using the Adams–Bashforth–Moulton Method

For numerical verification, the Adams–Bashforth–Moulton method was applied to Scenario S2 using the same initial conditions and scenario-specific parameter configuration employed in the preceding numerical analysis (Figure 7). The initial conditions were fixed at $D(0) = 0.90$, $H(0) = 0.45$, $S(0) = 0.75$, $R(0) = 0.45$, and $C(0) = 0.15$, and all model coefficients were fixed at the Scenario S2 values reported in Table 4. The fractional order was set to $\alpha = 0.80$.

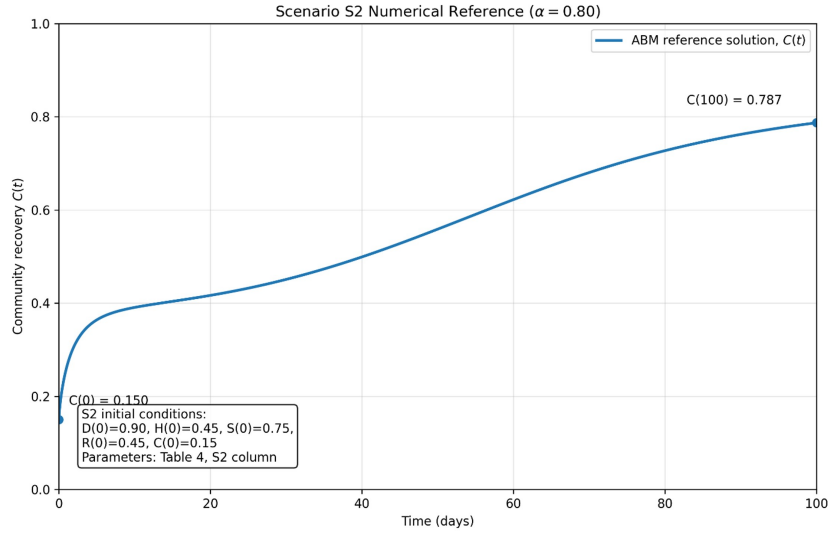


Figure 7. Adams–Bashforth–Moulton (ABM) numerical reference solution for community recovery $C(t)$ under Scenario S2 with fractional order $\alpha = 0.80$

Note: The initial conditions were fixed at $D(0) = 0.90$, $H(0) = 0.45$, $S(0) = 0.75$, $R(0) = 0.45$, and $C(0) = 0.15$, and all model parameters were fixed at the Scenario S2 values reported in Table 4. The simulation was performed over 100 days using a uniform time step of $\Delta t = 0.1$ day, yielding $C(100) = 0.787$.

The Adams–Bashforth–Moulton reference solution was computed over the post-earthquake interval $t \in [0, 100]$ days using a uniform time step of $\Delta t = 0.1$ day, resulting in 1001 temporal grid points. Under this configuration, the community recovery trajectory reached $C(100) = 0.787$. This numerical reference solution provides an internally consistent benchmark for the Scenario S2 fractional dynamics presented in the study. The Adams–Bashforth–Moulton solution presented serves as the numerical reference for the fractional dynamics considered in the present study. The physics-informed neural network formulation described in Section 6 remains an integral computational component of the proposed framework because the governing fractional equations and initial-condition constraints are embedded directly into its learning objective. However, following the numerical consistency check performed during revision, the present study does not make an independent quantitative accuracy claim for the physics-informed neural network surrogate. A systematic quantitative physics-informed neural network–Adams–Bashforth–Moulton benchmark across multiple parameter configurations is therefore reserved for future computational validation studies.

8.6 Comparative Numerical Outcomes

The comparative results (Table 5) indicate clear differences among the four synthetic scenarios. S4, characterized by stronger response and accelerated recovery conditions, produced the highest final recovery and the lowest final damage and shelter need. In contrast, S1 exhibited slower recovery under comparatively limited response conditions, while S3 produced the highest peak public health burden. These results illustrate how differences in response capacity, health burden, and scenario-specific parameter configurations influence the simulated recovery trajectories.

Table 5. Comparative recovery outcomes across scenarios (S1–S4)

Outcome	S1	S2	S3	S4
Final recovery $C(100)$	0.56	0.79	0.63	0.88
Peak health burden	0.80	0.71	0.93	0.55
Final damage $D(100)$	0.32	0.22	0.29	0.15
Final shelter need $S(100)$	0.37	0.24	0.31	0.18
Final response $R(100)$	0.47	0.74	0.64	0.82

8.7 Sensitivity Analysis

To evaluate the influence of memory effects on post-earthquake recovery, a sensitivity analysis was performed by varying the fractional-order parameter α from 0.60 to 1.00 while keeping all other parameters fixed. The results indicate that recovery trajectories are highly sensitive to α . Lower fractional-order values produced slower recovery and prolonged public health burden, whereas values closer to one generated faster recovery. These findings confirm that memory-dependent dynamics significantly influence post-earthquake recovery behavior. In addition to the fractional-order parameter- α , the model structure suggests that several interaction coefficients may have a substantial influence on the recovery trajectory. In particular, γ_R , which controls the positive contribution of response capacity to community recovery, is expected to directly affect the rate of recovery. The parameters γ_D and γ_H , representing the negative effects of damage intensity and public health burden on community recovery, respectively, are also expected to influence both the speed and final level of recovery. Similarly, β_R , β_H , and β_S determine the effectiveness of response capacity in reducing damage, public health burden, and shelter need and therefore may indirectly affect the recovery trajectory.

These observations are based on the mathematical structure of the governing equations and should not be interpreted as a global parameter-ranking analysis. A comprehensive variance-based or global sensitivity analysis will be considered in future applications of the framework using empirically calibrated parameters. Because all five state variables were normalized to the interval $[0,1]$, the coefficients used in the present scenario-based simulations were treated as scaled, dimensionless parameters within the normalized computational framework. Thus, the reported coefficient values represent relative interaction strengths rather than dimensional physical rates. This scaling provides consistency across the coupled equations and facilitates comparison among the four synthetic scenarios. In future applications using dimensional empirical data, the coefficients and fractional-order terms will require calibration with units consistent with the selected time scale and the fractional derivative formulation.

9 Discussion

The findings of this study show that earthquake recovery is a nonlinear and memory-dependent process that depends on the memory of the earthquake and its effects. The numerical results presented demonstrate that earthquake recovery is dependent on more than just the damage that the earthquake causes initially but also on the public health burden and the number of individuals who require shelter after the earthquake. These findings support the main assumption of this proposed framework. Another finding of this research study is the importance of the public health burden after earthquakes. Most disaster models do not include the impact of the health of the population after those disasters. However, by including the public health burden as a variable in the earthquake recovery model, it becomes evident that the health of the population after earthquakes directly affects the recovery of those affected areas. For earthquake-prone regions of the world, this is a particularly important finding. These findings are consistent with recent studies emphasizing resilience-based recovery planning and integrated disaster-health frameworks [27–30].

Further analysis of the fractional-order system shows that memory effects are important in the recovery of the area post-earthquake. The lower fractional-order values indicated slower recovery of the area post-earthquake, suggesting that the effects of the earthquake and the recovery efforts made during that period had an impact upon the ability of the area to recover in the future. Thus, fractional calculus can be utilized to model the recovery of the area post-earthquake. From a policy perspective, the proposed framework may support emergency managers and public health authorities in identifying critical periods of recovery. For example, knowledge of the trajectories of the illness and the effects of memory upon those who experienced it could enable authorities to more effectively plan for the need for resources during those periods, such as medical resources, shelter, and the restoration of critical infrastructure.

The trajectory of each of these model variables can provide some insight into the needs of the earthquake victims. The variable $R(t)$ indicates the level of response that is occurring to the earthquake victims. Low or increasing values of this variable indicate periods during which the response resources (emergency personnel, medical personnel and resources, logistics, transportation, etc.) were insufficient to respond to the earthquake victims. High and rapid increases in this variable indicate periods during which the implementation of these response resources was rapidly leading to the reduction of the earthquake's damaging effects on the earthquake victims. The shelter need variable $S(t)$ indicates the length and severity of the need for temporary shelters for earthquake victims. High values of this model variable that slowly decline with time indicate periods during which there was a need for temporary shelters and services for the earthquake victims. Thus, the variable $S(t)$ could be utilized in future models, which incorporate empirical data from past earthquakes, to determine which strategies for establishing temporary shelters would lead to the best outcomes for earthquake victims.

Similarly, the variable $H(t)$ indicates the level of strain that the public health system of the region was experiencing as a result of the earthquake. High values of this model variable indicate periods during which the healthcare resources of the region were overwhelmed by the demands of the earthquake victims. Thus, by modeling both the health system strain $H(t)$ and the response system resources $R(t)$, it is possible to indicate periods during which additional health system resources would have had the greatest potential value. These interpretations also have implications for the

sustainable recovery from earthquakes. The mobilization of resources is only one component of the earthquake response; coordinating the response with the shelter, healthcare, and infrastructure recovery and restoration efforts can help to alleviate the dependence upon emergency resources and contribute to the development of more sustainable and resilient recovery pathways for earthquake-affected communities. However, because these results are based upon the synthetic scenario of earthquakes, rather than upon earthquake observations and data, these interpretations should not currently be applied to earthquake response and management efforts. Validation with real post-earthquake data is required before the framework can be used for operational resource allocation.

Subject to future empirical calibration and validation, the proposed framework may support the evaluation of emergency-response strategies related to resource allocation, preparedness, shelter planning, and public health response. Additionally, the model could be extended to incorporate real earthquake records, public health records of hospitalizations, and various measures of public health recovery. Another limitation of the present framework is the use of a common fractional-order parameter for all state variables. In reality, damage intensity, public health burden, shelter demand, response capacity, and community recovery may evolve according to different memory scales. Future studies may therefore consider multi-order fractional systems in which each state variable is governed by its own fractional-order parameter.

The limitations of this study are mainly in the fact that the data used was scenario-driven and normalized data rather than empirical data from the earthquake patterns of the region. Additionally, the model has yet to incorporate elements like spatial aspects of the region, uncertainty of earthquake outcomes, or the impact of other regions within the population. Future studies could extend this model by incorporating real earthquake data, hospital admissions data, displacement data, mobility data, and Geographic Information Systems (GIS) data relating to the regions of interest.

10 Conclusion

A physics-informed fractional framework was proposed to model disaster risk, public health burden, shelter needs, response capacity, and the recovery process following an earthquake. The model uses a system of fractional differential equations to represent memory-dependent interactions among the principal components of post-earthquake recovery. Scenario-based numerical simulations and the Adams–Bashforth–Moulton reference solution were used to examine the resulting system dynamics. The framework additionally incorporates a physics-informed neural network formulation in which the governing fractional equations and initial-condition constraints are embedded into the computational learning objective. In the present analysis, numerical consistency of the scenario-based dynamics was assessed using the Adams–Bashforth–Moulton reference solution, while broader quantitative validation of the physics-informed neural network component remains a direction for future work. The scenario-based numerical simulations indicate that damage intensity, public health burden, shelter need, and response capacity can influence the simulated recovery trajectories. Following empirical calibration and validation, the proposed framework may have potential applications in disaster preparedness, public health planning, and post-earthquake recovery assessment. Future work will involve implementing this model into real earthquake data to evaluate its accuracy, as well as extending the model to incorporate additional factors. These findings highlight the importance of integrating public health, infrastructure, and response capacity into a unified disaster recovery framework.

Author Contributions

Conceptualization, A.Y. and N.Y.; methodology, A.Y.; software, A.Y. and S.E.A.; validation, A.Y. and S.E.A.; formal analysis, A.Y.; investigation, A.Y. and N.Y.; resources, A.Y. and N.Y.; data curation, A.Y. and S.E.A.; writing—original draft preparation, A.Y. and N.Y.; writing—review and editing, A.Y., N.Y. and S.E.A.; visualization, A.Y. and S.E.A.; supervision, A.Y.; project administration, A.Y. All authors have read and agreed to the published version of the manuscript.

Data Availability

The synthetic scenario-based data and parameter configurations supporting the findings of this study are included within the article. No external empirical dataset was used in the present study. Additional information related to the computational implementation is available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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